

IUBAT - INTERNATIONAL UNIVERSITY OF BUSINESS AGRICULTURE AND TECHNOLOGY

School of Computer Science and Engineering | Department of CSE

Mid Term Examination " Summer Semester 2026

Course Code: CSE 4385 | Course Title: Computer Vision and Image Processing

Time: 01 Hour 30 Minutes | Full Marks: 100

Instructions: Answer all questions. Each question carries 10 marks (10 x 10 = 100).

1. An 8-bit grayscale image has a discrete dynamic range of pixel intensities spanning: [CO2, C1, PO1 - 10 Marks]

☐ (A) 0 to 127

☐ (B) 0 to 255

☐ (C) 1 to 256

☐ (D) -128 to 127

2. In digital image processing, what is the primary objective of Histogram Equalization? [CO1, C2, PO1 - 10 Marks]

☐ (A) To compress the spatial resolution of an image

☐ (B) To adjust contrast by flattening intensity distribution

☐ (C) To convert a 3-channel RGB image into binary

☐ (D) To reduce Gaussian noise using spatial averaging

3. If an RGB pixel has normalized coordinates (R=1.0, G=0.0, B=0.0), its CMY representation is: [CO2, C3, PO1 - 10 Marks]

☐ (A) (C=0.0, M=1.0, Y=1.0)

☐ (B) (C=1.0, M=0.0, Y=0.0)

☐ (C) (C=0.5, M=0.5, Y=0.0)

☐ (D) (C=0.0, M=0.0, Y=0.0)

4. In the HSI color model, which parameter represents the purity or vibrancy of a color? [CO2, C1, PO1 - 10 Marks]

☐ (A) Hue

☐ (B) Saturation

☐ (C) Intensity

☐ (D) Luminance

5. A binary image requires how many bits to store each pixel? [CO1, C2, PO1 - 10 Marks]

☐ (A) 1 bit

☐ (B) 8 bits

☐ (C) 16 bits

☐ (D) 24 bits

6. Which image file format uses lossless compression and supports alpha-channel transparency? [CO1, C1, PO1 - 10 Marks]

☐ (A) JPEG

☐ (B) BMP

☐ (C) PNG

☐ (D) GIF

7. Which spatial domain operation is primarily used for edge detection by approximating gradient magnitude? [CO1, C3, PO1 - 10 Marks]

☐ (A) Sobel Operator

☐ (B) Median Filter

☐ (C) Histogram Matching

☐ (D) Bit-Plane Slicing

8. Morphological Dilation using a horizontal rectangular kernel (50x10) is applied in OCR pre-processing to: [CO1, C3, PO1 - 10 Marks]

☐ (A) Shrink foreground objects

☐ (B) Bridge gaps between letters to form continuous lines

☐ (C) Invert grayscale gradients

☐ (D) Sharpen fine diagonal stroke edges

9. In the digital image processing pipeline, what is the key function of 'Feature Selection'? [CO1, C2, PO1 - 10 Marks]

☐ (A) Digitizing analog signals into discrete arrays

☐ (B) Retaining non-redundant quantitative attributes

☐ (C) Equalizing background color distribution

☐ (D) Removing salt-and-pepper noise

10. Otsu's thresholding algorithm calculates an optimal global threshold value by: [CO1, C4, PO1 - 10 Marks]

☐ (A) Maximizing inter-class variance between background and foreground

☐ (B) Taking arithmetic mean of max and min pixels

☐ (C) Computing fast Fourier transform of gradients

☐ (D) Applying random sampling across 3x3 patches